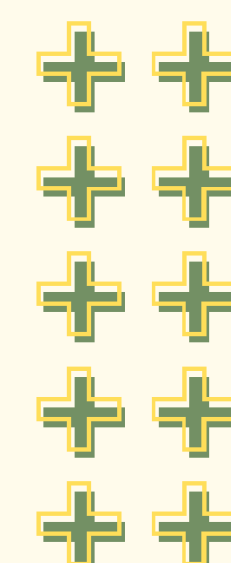
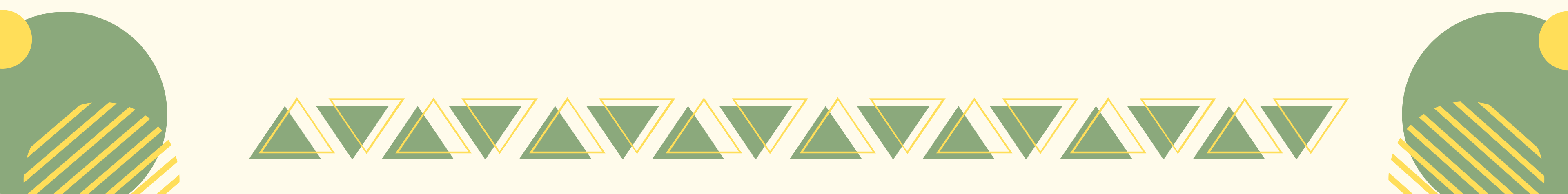


Uncovering High-Risk Indicators and Transaction Behavior



by Radhwaa Nur Naa'ila



Dataset Overview & Class Imbalance

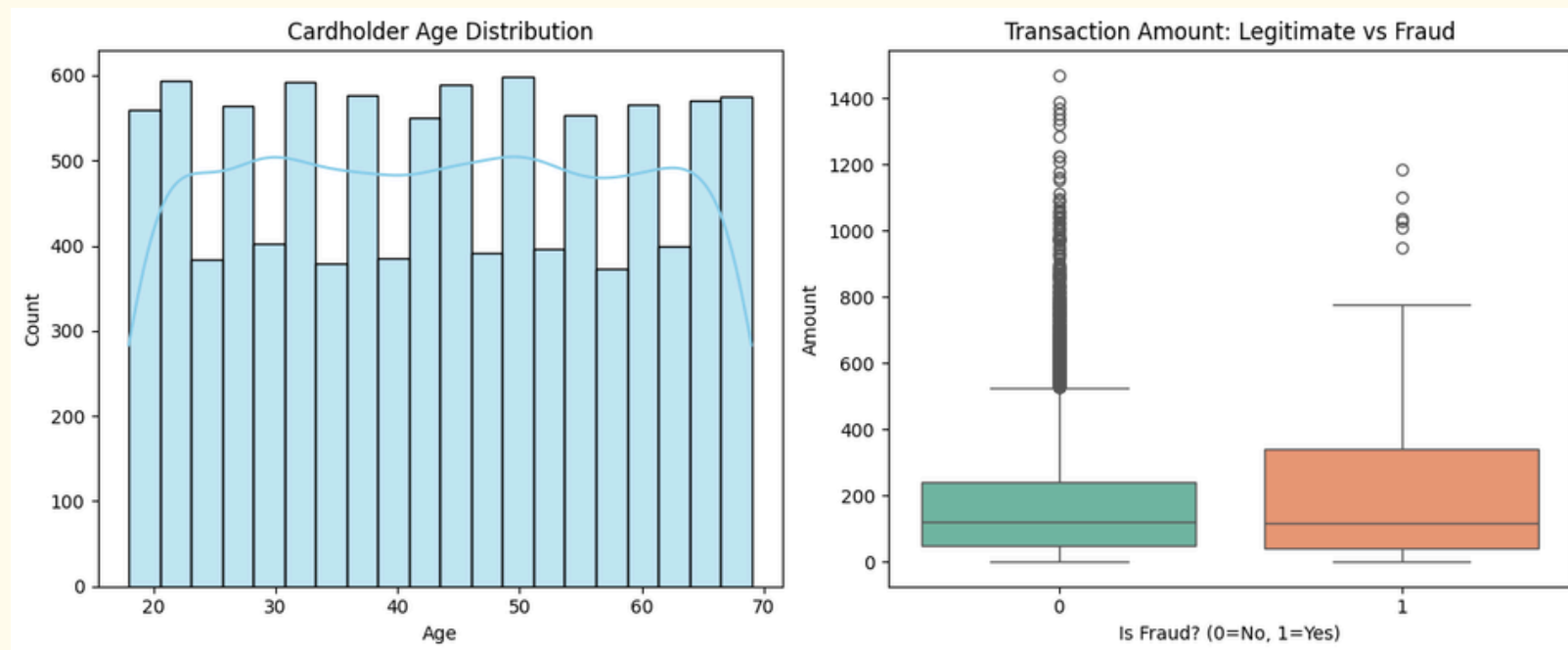
```
--- DATASET INFORMATION ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  -
0   transaction_id       10000 non-null  int64
1   amount              10000 non-null  float64
2   transaction_hour     10000 non-null  int64
3   merchant_category   10000 non-null  object
4   foreign_transaction  10000 non-null  int64
5   location_mismatch   10000 non-null  int64
6   device_trust_score  10000 non-null  int64
7   velocity_last_24h   10000 non-null  int64
8   cardholder_age      10000 non-null  int64
9   is_fraud             10000 non-null  int64
dtypes: float64(1), int64(8), object(1)
memory usage: 781.4+ KB

--- STATISTICAL SUMMARY ---
      transaction_id      amount  transaction_hour  foreign_transaction \
count      10000.00000      10000.00000      10000.00000      10000.00000
mean         5000.50000      175.949849         11.593300         0.097800
std         2886.89568      175.392827         6.922708         0.297059
min           1.00000         0.000000         0.000000         0.000000
...
is_fraud
0      9849
1       151
Name: count, dtype: int64

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
C:\Users\Lenovo\AppData\Local\Temp\ipykernel_9900\4181964013.py:31: FutureWarning:
```

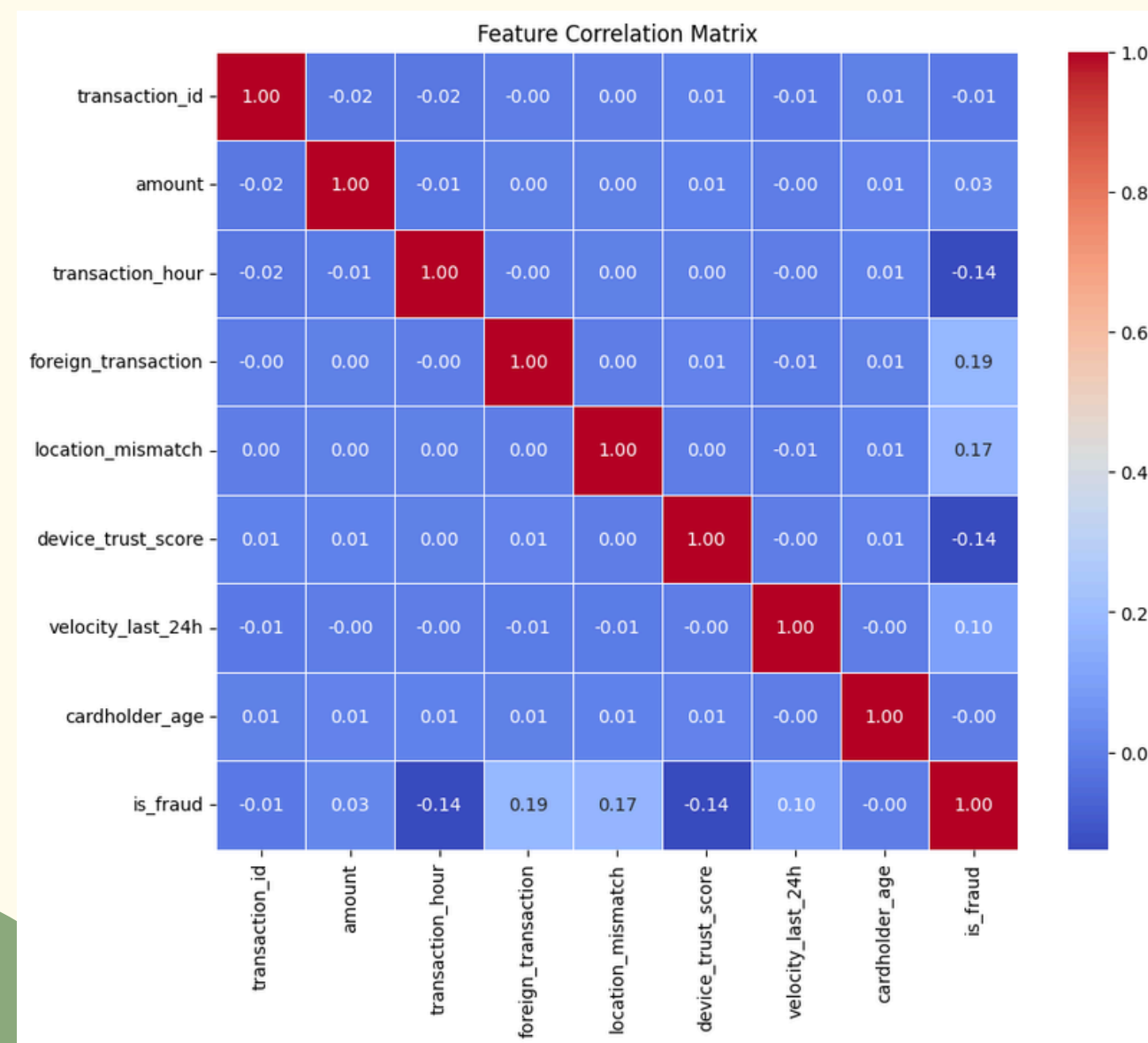
- Dataset Profile: 10,000 transaction records | 10 core columns | Zero missing values.
- Class Target Distribution (is_fraud): * Legitimate Transactions (0): 9,849 cases (98.49%)
- Fraudulent Transactions (1): 151 cases (1.51%)
- Key Takeaway: The data shows a severe class imbalance, which is a standard characteristic of financial anomaly detection.

Cardholder Age Distribution Analysis



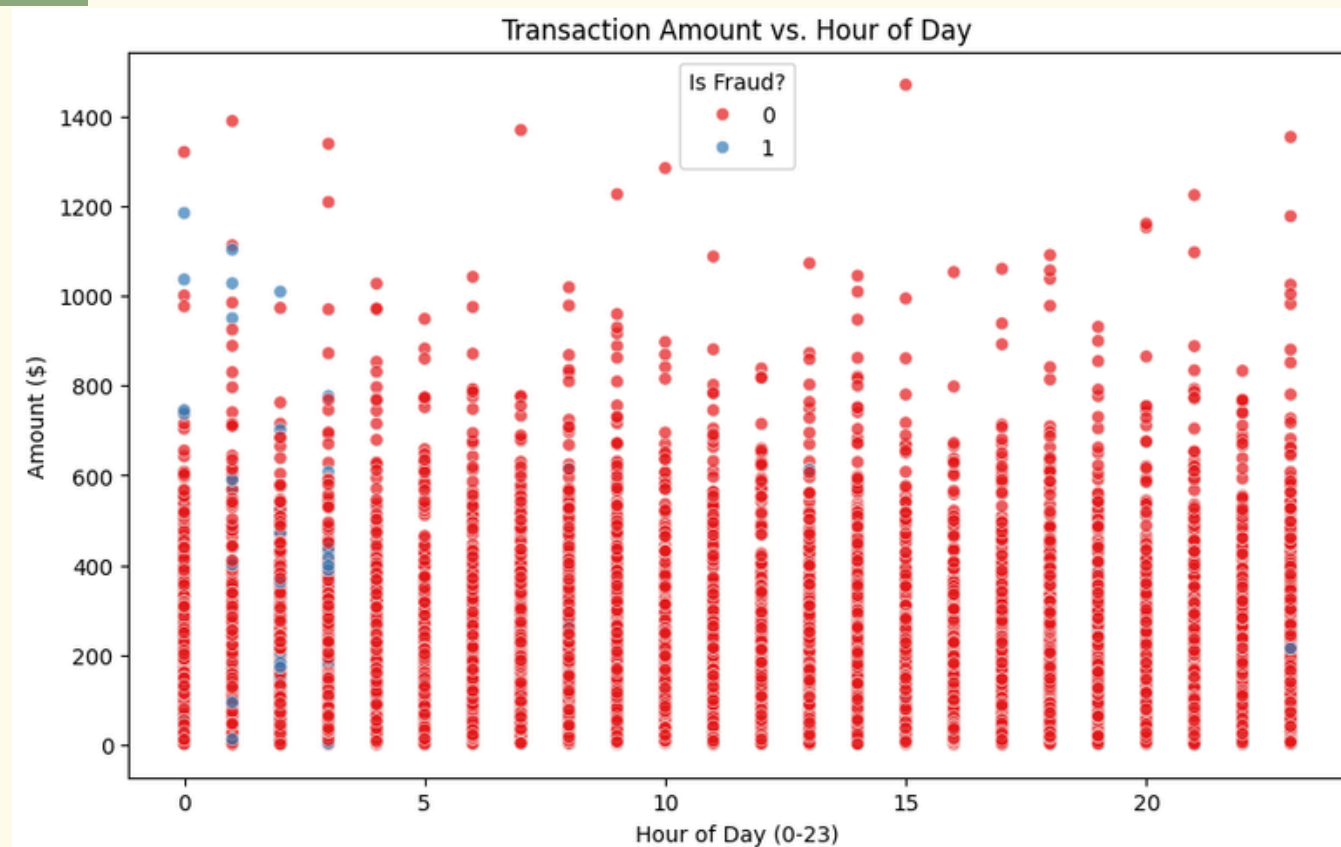
- **Demographic Exploration:** The histogram shows an identical bell-curve distribution for both classes.
- **Statistical Insight:** The correlation between `cardholder_age` and `is_fraud` is near-zero (-0.00059).
- **Model Optimization:** Demographic age holds no predictive power for identifying fraud and can be removed during feature engineering to eliminate noise.

Feature Correlation Matrix



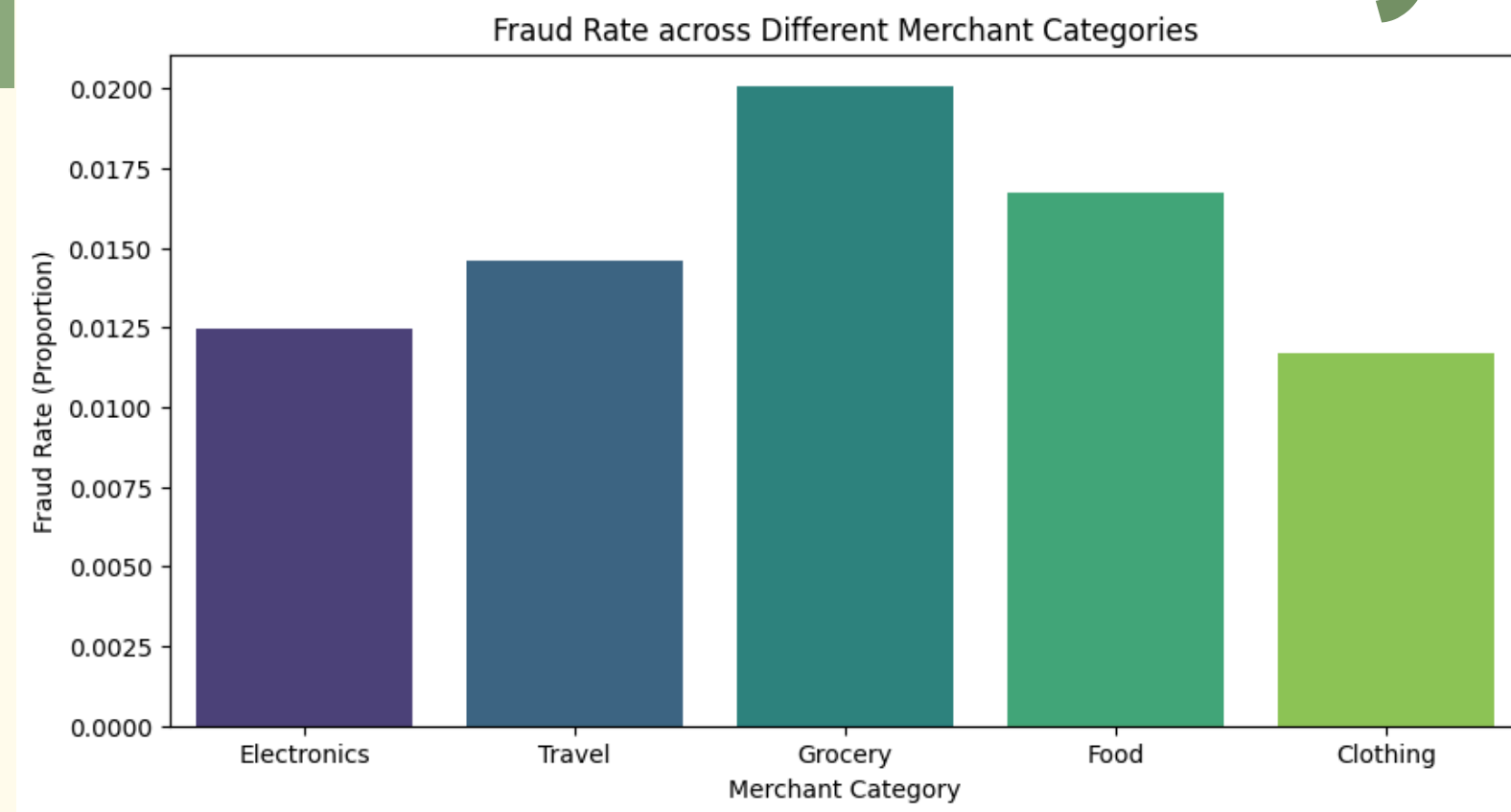
- **Top Positive Drivers:** foreign_transaction (0.19) and location_mismatch (0.17) have the strongest linear relationship with fraud.
- **Top Negative Drivers:** transaction_hour (-0.14) and device_trust_score (-0.14) indicate inverse risk factors.
- **Security Flag:** Transactions utilizing untrusted hardware outside the home country present exponential risk.

Behavioral Analysis: The "Night Owl" Pattern

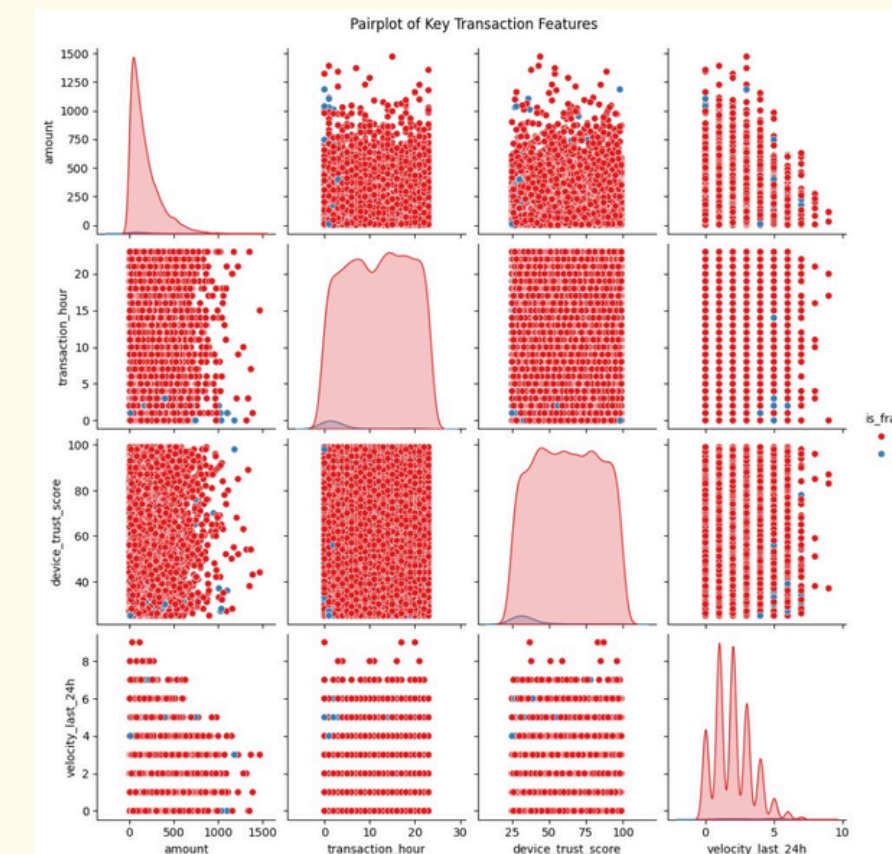


- **Temporal Distribution:** Clear clustering of fraudulent events (red/alternative color markers) during early morning windows.
- **Key Statistics:** Fraud peaks heavily around 3:41 AM, while legitimate client traffic is centered at 11:41 AM.
- **Behavioral Insight:** Time-of-day execution is a more critical separator for fraud than the transaction size alone.

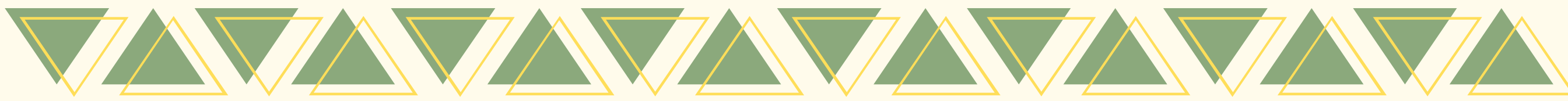
Transaction Value & Industry Risk Sector



- **Financial Profile:** Fraudulent transactions exhibit higher average values (\$216.18) compared to standard checkouts (\$175.33).
- **Sector Vulnerability:** The Grocery sector carries the highest fraud rate (2.0%), followed by Food (1.67%) and Travel (1.45%).
- **Operational Threat:** Rapid velocity spikes combined with high ticket sizes in grocery categories suggest immediate system exploitation.



- **Channel & Sector Risk Analysis:**
- **Highest Risk Sector:** Grocery stores exhibit the highest fraud proportion at approximately 2.0%.
- **Secondary Targets:** Food (1.67%) and Travel (1.45%) closely follow as primary exploitation channels.
- **Lowest Risk Sector:** Clothing shows the minimum vulnerability to fraudulent activities.
- **Operational Insight:** High-frequency transaction velocity targeted at Grocery or Food merchants should be flagged by the system as high-risk behavior.



Conclusion & Actionable Security Recommendations

Data-Driven Security Framework:

1. **Dynamic Authentication:** Trigger mandatory Multi-Factor Authentication (OTP) for transactions between 12:00 AM and 5:00 AM if a `location_mismatch` occurs.
 2. **Device Rules:** Instantly decline international routing requests if the `device_trust_score` falls under a critical threshold of 40.
 3. **Velocity Caps:** Implement account cooling periods if user transactions exceed 3 sessions within 24 hours in the grocery sector.
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Thank you